

Donkron Seminar #3: Uhlenbeck's Thm 1 - Motivation + Statement

Upshot Uhlenbeck's thm gives existence of local coulomb gauges

Recall Lecture #1:

We want solutions to the ASD equation
(ASD) instantons

$\Omega_x^2(E)$ splits as $\Omega_x^2(E) = \underbrace{\Omega_x^+(E)}_{+ \text{ eigenspace}} \oplus \underbrace{\Omega_x^-(E)}_{- \text{ eigenspace}}$
of Hodge $\star$

Since $F_A \in \Omega_x^2$, it also splits. $F_A = F_A^+ \oplus F_A^-$

If $F_A^+ = 0$, then $F_A \in \Omega_x^-$, i.e., it is entirely anti-self dual

We call A s.t. $F_A^+ = 0$ ASD Instantons.

Issue #1 Solutions with many faces

Recall that Gauge transformations act on connections by:

Bundle Automorphisms $\nabla_{u(A)} s = u \nabla_A (u^{-1} s) = \nabla_A s - (\nabla_A u) u^{-1} s$

Locally, this action is $u(A) = u A u^{-1} - (du) u^{-1}$ on connections

$F_{u(A)} = u F_A u^{-1}$ on curvature

Since $\star$ and u act separately,

$$u (F_A^+ + F_A^-) u^{-1} = u F_A^+ u^{-1} + u F_A^- u^{-1}$$

So if $F_A^+ = 0$, $u F_A^+ u^{-1} = 0$

i.e., get a gauge gp's worth of solⁿ for every solⁿ. Ω_x^+

But recall, $u(A) = u A u^{-1} - u^{-1} du$

In particular, for $A \equiv 0$ trivial, $u(0) = -u^{-1} du$

$$\text{And } \|A\|_{L^2} = 0 \nRightarrow \|u(A)\| = \|u^{-1} du\|_{L^2} = 0.$$

In fact, u can be as large as possible, since u is only C^∞ ; we have no control on its derivatives

Q Can we restrict to some subset of Ω_x^1 to ensure that

- (i) there is only one representative of each Gauge equivalence class
- (ii) that representative is sufficiently nice? i.e. is in L^2

Ensuring something like this is typically called "gauge fixing" or "choosing a gauge slice" (i.e. Coulomb gauge)

Issue #2 How to actually find solⁿs to ASD?

We can rewrite $F_A^+ = \underbrace{d^+(A)}_{\text{non-elliptic}} + \underbrace{(A \wedge A)^+}_{\text{non-linear}} = 0$ in local coordinates

$$d^+ : \Omega^1 \xrightarrow{d} \Omega^2 \xrightarrow{P_+} \Omega^2_+$$
$$P_+ = \frac{1}{2}(1 + *)$$

This is a non-linear, non-elliptic PDE!

Q Can we replace it with something linear + elliptic?

Step 1) Linearize: consider solⁿs of $d^+(A) = 0$

Still not elliptic!

Why? compute principle symbol, or note that

so any A st $d(A) \in \Omega^-$ is in the kernel of d^+

Step 2) Make elliptic.

Note that for any connection $\tilde{A}$ with $d^*\tilde{A} = 0$, ↖ define later

the operator $d^* + d : \Omega_x^1 \rightarrow \Omega_x^0 \oplus \Omega_x^2(E)$

is elliptic, by the splitting $\Omega_x^1 = H^1 \oplus \text{im } d^* \oplus \text{im } d$

$$\Rightarrow \ker d^* \subset \text{im } d \oplus H^1$$

$\Rightarrow$ if $H^1(X) = 0$, $\Omega_X^1 = \text{im } d^* \oplus \text{im } d$, so $\ker d^* \subset \text{im } d$

$\Rightarrow$ every 1-form is $\perp$ to $\ker(d+d^*) \Rightarrow \tilde{A} \in \text{im } d$

Recall from last week: Since $d+d^*$ is elliptic,

$$\|\tilde{A}\|_{L^2_k} \leq C \left(\|d\tilde{A}\|_{L^2_{k-1}} + \|d^*\tilde{A}\|_{L^2_{k-1}} \right) \quad (\text{elliptic regularity})$$

So if $d^*\tilde{A} = 0$,

$$\|\tilde{A}\|_{L^2_k} \leq C \|F\tilde{A}\|_{L^2_{k-1}}$$

Upshot: For every $\tilde{A}$ s.t. $d^*\tilde{A} = 0$, if we also want $d_+\tilde{A} = 0$,

we can replace ASD eqnⁿ with $\underbrace{(d^+ + d^*)(A)}_{\text{this is elliptic!}} = 0$

Q So, can we actually do this?

A, Yes, for A with small enough curvature on triv. bundle.

Thm: [Uhlenbeck]

$\exists$ constants ε, M s.t. any connection A

on the trivial bundle over $\overline{B^4}$ s.t. $\|F_A\|_{L^2} < \varepsilon$

is gauge equivalent to $\tilde{A}$ over B^4 s.t.

- $d^*\tilde{A} = 0$
 - $\lim_{r \rightarrow 1} \tilde{A}_r = 0$
 - $\|\tilde{A}\|_{L^2_1} \leq C \|F_A\|_{L^2}$ have to control $|d/dx|$
- $$\|\tilde{A}\|_{L^2_1} = \left(\int_{B^4} |\nabla \tilde{A}|^2 + |\tilde{A}|^2 d\mu \right)^{1/2}$$

Even more: if ε, C are chosen "suitably",

$\tilde{A}$ is determined uniquely up to constant gauge

transformation: $\tilde{A} \mapsto u_0 \tilde{A} u_0^{-1}$

Cor $\exists \varepsilon > 0$ s.t. $\forall$ sequence $\{A_i\}$ of ASD connections over $\overline{\mathbb{D}^4}$

s.t. $\|F(A_i)\|_{L^2} \leq \varepsilon \forall i$, $\exists$ a subsequence $\{j\} \subset \{i\}$ and

gauge equivalent connections $\tilde{A}_j$ s.t. $\{\tilde{A}_j\}$ converge ^(C^∞) in $\mathbb{D}^4$

For the physicists:

Yang-Mills action functional:

Set $\text{Conn}(E) = \text{connections on } E$. for $\begin{matrix} E \\ \downarrow \\ (M, g) \end{matrix}$

Then $YM(A) = \int_M |F_A|^2 d\text{vol}(g)$

$F_A = \text{curvature}$.

this is action/energy integral, and we want to find minimizers.

Perturb A by $a \in \mathfrak{su}(2)$, plug in $A + ta$:

$$\begin{aligned} F_{A+ta} &= d(A+ta) + (A+ta) \wedge (A+ta) \\ &= dA + A \wedge A + t d_A a + t^2 a \wedge a \\ &= F_A + t d_A a + \underbrace{t^2 a \wedge a}_{\text{dies for } t \ll 1} \end{aligned}$$

$$\|F_{A+ta}\|^2 = \|F_A\|^2 + 2t \langle d_A a, F_A \rangle + \mathcal{O}(t^2)$$

Then differentiate at 0:

$$\begin{aligned} \left. \frac{d}{dt} YM(A+ta) \right|_{t=0} &= \left. \frac{d}{dt} \right|_{t=0} 2 \int_M \langle d_A a, F_A \rangle d\text{vol}(g) \\ &= 2 \int_M \langle a, d_A^* F_A \rangle d\text{vol}(g) \end{aligned}$$

To get Euler-Lagrange, let 1st variation to 0 $\forall a$.

$$2 \int_M \langle a, d_A^* F_A \rangle d\text{vol}(g) = 0 \quad \forall a \Rightarrow d_A^* F_A = 0$$

Claim: Solⁿs to $d_A^* F_A = 0$ are ASD instantons